

AIMS System - Detailed Class Design Documentation

17. Design Class: DeliveryInfo

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	recipientName	String	null	Name of the recipient
2	email	String	null	Email address
3	phoneNumber	String	null	Phone number
4	province	String	null	Province/city
5	district	String	null	District
6	address	String	null	Delivery address
7	deliveryMethod	String	"Regular"	Delivery method (Regular/Rush)
8	deliveryTime	String	0.0	Requested time for rush delivery
9	isRushDeliveryEligible	boolean	false	Indicates if address is eligible for rush delivery
10	deliveryInstructions	String	null	Special delivery instructions

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	DeliveryInfo	void	Constructor for creating delivery information
2	validateDeliveryInfo	boolean	Validates all required delivery information fields
3	isEligibleForRushDelivery	boolean	Calculates shipping fee based on product weight and location
4	calculateShippingFee	float	Calculates shipping fee based on product weight and location
5	calculateRushDeliveryFee	float	Calculates rush delivery fee for eligible products
6	getDeliveryInfo	Map<String, String>	Returns all delivery information as a map
7	updateDeliveryInfo	boolean	Updates the delivery information with new values
8	isInHanoiInnerCity	boolean	Checks if delivery is to inner Hanoi districts

Parameter:

- **DeliveryInfo(recipientName: String, email: String, phoneNumber: String, province: String, district: String, address: String, deliveryMethod: String)**
 - **recipientName:** Full name of the recipient
 - **email:** Contact email address
 - **phoneNumber:** Contact phone number
 - **province:** Delivery province/city
 - **district:** Delivery district
 - **address:** Detailed delivery address
 - **deliveryMethod:** Delivery method (Regular or Rush)
- **validateDeliveryInfo(): boolean**
 - **Return:** true if all required fields are valid, false otherwise
- **isEligibleForRushDelivery(): boolean**
 - **Return:** true if the address is in an area that supports rush delivery, false otherwise
- **calculateShippingFee(weight: float): float**
 - **weight:** Total weight of the products in kilograms
 - **Return:** The calculated shipping fee in VND
- **calculateRushDeliveryFee(numItems: int): float**
 - **numItems:** Number of items for rush delivery
 - **Return:** The calculated rush delivery fee in VND
- **getDeliveryInfo(): Map<String, String>**
 - **Return:** Map containing all delivery information fields
- **updateDeliveryInfo(updatedInfo: Map<String, String>): boolean**

- **updatedInfo**: Map containing updated delivery information fields
 - **Return**: true if update was successful, false otherwise
- **isInHanoiInnerCity(): boolean**
 - **Return**: true if the delivery district is in inner Hanoi city, false otherwise

Exception:

- **validateDeliveryInfo(): boolean**
 - **InvalidEmailException**: Thrown if email format is invalid
 - **InvalidPhoneNumberException**: Thrown if phone number format is invalid
 - **MissingRequiredFieldException**: Thrown if any required field is missing
- **calculateShippingFee(weight: float): float**
 - **InvalidWeightException**: Thrown if weight is negative or zero
 - **UnsupportedLocationException**: Thrown if delivery location is not supported

Method:

- **validateDeliveryInfo(): boolean**
 - **Purpose**: Validates all required delivery information fields for completeness and format
 - **Processing**:
 1. Checks that recipient name is not empty
 2. Validates email format using regex pattern
 3. Validates phone number format using regex pattern
 4. Checks that province, district, and address are not empty
 5. If delivery method is "Rush", checks that the address is eligible for rush delivery
 - **Return**: true if all validations pass, false otherwise with appropriate error messages
- **isEligibleForRushDelivery(): boolean**
 - **Purpose**: Determines if the delivery address is eligible for rush delivery
 - **Processing**:
 1. Checks if province is "Hanoi"
 2. Checks if district is one of the inner city districts of Hanoi
 - **Return**: true if in eligible area, false otherwise
- **calculateShippingFee(weight: float): float**
 - **Purpose**: Calculates the shipping fee based on product weight and delivery location
 - **Processing**:
 1. Checks if delivery is to inner city Hanoi or Ho Chi Minh City
 - If yes, base fee is 22,000 VND for first 3kg
 - If no, base fee is 30,000 VND for first 0.5kg
 2. Adds 2,500 VND for every additional 0.5kg
 3. Applies free shipping if order value exceeds 100,000 VND (up to 25,000 VND)
 - **Return**: The calculated shipping fee
- **calculateRushDeliveryFee(numItems: int): float**
 - **Purpose**: Calculates additional fee for rush delivery items
 - **Processing**:
 1. Multiplies number of rush delivery items by 10,000 VND
 2. Adds this to the base shipping fee
 - **Return**: The calculated rush delivery fee

State:

- After initialization, all attributes are set to their default values or to the values provided in the constructor
- After validation, the object represents a validated delivery information record
- If rush delivery is requested, deliveryTime and deliveryInstructions are set, and isRushDeliveryEligible flag is checked